

AISSAOUI Ismail

AI & Data Science Engineer

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PROFILE

AI & Data Science Engineer passionate about human-computer interaction and adaptive learning systems. Expertise in Transformer models, Reinforcement Learning, and Digital Twins. PhD candidate for co-learning in XR, aiming to merge embodied AI and technical gesture simulation for advanced training environments.

SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL

AI / ML: PyTorch, TensorFlow, Keras, Scikit-learn, **Inverse RL**, **Imitation Learning**

AI Techniques: Transformers, GNN, GANs, RL (DQN, PPO), **Symbolic AI** (Rule-based systems)

Optimization: PSO, Genetic Algorithm, ACO, Simulated Annealing, Bayesian Optimization

Web, API & 3D: Next.js, FastAPI, Node.js, **Unity3D**, **Unreal Engine (C++)**

Data & MLOps: MLflow, Airflow, Kubeflow, DVC, Pandas, NumPy

Databases: PostgreSQL, MongoDB, MySQL, Neo4j, Cassandra

Cloud & DevOps: Docker, Linux, Git, CI/CD, CUDA, AWS, Grafana

Soft Skills: Technical leadership, Communication, Stakeholder management

PROFESSIONAL EXPERIENCE

AI Engineer Intern — Digital Twin & Simulation

2026

Sonatrach — Algeria

Developed a Digital Twin for real-time supervision of gas turbines. Modeled physical flows and estimated RUL using Deep Learning. This immersion in high-fidelity simulation enabled me to master the digital replication of complex industrial processes, a cornerstone for immersive training systems.

Network Engineering Intern

Sep 2024

Algérie Télécom RMC Center — Chlef, Algeria

EDUCATION

National Higher School of Computer Science (ESI-SBA)

2021–2026

Engineering and Master's Degree in Computer Science

Specialization: Artificial Intelligence & Informatics

KEY ACHIEVEMENTS

- Developed custom Transformer models (BERT, GPT, RAG) for real-world multilingual applications.

- Built explainable AI platforms for HR tech and public institutions — reducing bias and improving decision quality.
- Designed a graph-based recommendation system with +24% accuracy improvement (dataset: 60K profiles).
- Delivered complete ML pipelines: data ingestion, validation, training, CI/CD, and monitoring.
- Built real-time vision systems (YOLO, ByteTrack) reducing latency by 32%.

PROJECTS

Aura — Adaptive Reinforcement Learning System

2025

Developed an autonomous agent using **Reinforcement Learning** (DQN) for real-time strategy optimization. Implemented adaptive policies allowing the system to continuously learn from its environment.

Tech: TensorFlow, RL, Python — Concepts: Reinforcement Learning, Adaptability

Vision Transformer (ViT) Implementation

2025

Adapted Transformer architectures for Computer Vision. Treated images as sequences of patches, using multi-head self-attention to capture global dependencies.

Tech: PyTorch, Deep Learning — Concepts: Transformers, Image-to-Sequence

Recognizini — Online Learning & Interactive Feedback

2025

Real-time facial recognition system integrating human feedback loop for incremental identity learning. Used **Label Propagation** to dynamically enrich the model.

Tech: PyTorch, OpenCV, FastAPI — Concepts: Ecological Learning, Feedback Loop

Traffic Analytics & Safety Monitoring (YOLOv8)

2025

Analyzed video data streams for object tracking and trajectory estimation. Provides a solid foundation for modeling and evaluating **technical gestures** in real-time.

Tech: YOLOv8, Computer Vision — Concepts: Motion Analysis, Action Tracking

GNN-based Recruitment Intelligence

2024–2025

Modeled complex relationships between skills and profiles using **Graph Neural Networks**. Expertise in extracting rich semantic representations from knowledge graphs.

Tech: PyTorch Geometric, SBERT — Concepts: Graph Learning, Knowledge Modeling

Mini-GPT & Deep NLP Adaptation

2025

Implemented GPT (Decoder-only) architectures and LoRA adaptation for BERT models on multilingual and dialectal sentiment analysis tasks.

Tech: PyTorch, LoRA, HuggingFace — Concepts: LLM, Fine-tuning, NLP

Geo-RAG & Intelligent Scheduling

2025

Developed geospatial RAG systems and metaheuristic algorithms (GA, PSO) for solving complex NP-complete constraint optimization problems.

Tech: ChromaDB, LangChain, NumPy — Concepts: RAG, Optimization

Elliptic++ : Bitcoin Fraud Detection

2025

Detected illicit actors on the Bitcoin network via graph-based learning on over 200k transactions.

Tech: PyTorch Geometric, NetworkX — Concepts: Fraud Detection, GNN

LANGUAGES

Arabic: Mother Tongue | English: Advanced | French: Conversational